



Securing understanding of ratio as a multiplicative relationship

Task A: Reviewing ratio

1a) Construct a ratio table for the following recipes.

1 vanilla cake
250 g of flour
2 eggs
10 ml of vanilla
75 g of sugar
75 g of butter

Cake	Flour (g)	Eggs	Vanilla (ml)	Sugar (g)	Butter (g)
1	250	2	10	75	75
4	1000	8	40	300	300

$$1000 \div 250 = 4$$

1 banana cake
200 g of flour
2 eggs
3 bananas
200 g of sugar
200 g of butter

Cake	Flour (g)	Eggs	Bananas	Sugar (g)	Butter (g)
1	200	2	3	200	200
5	1000	10	15	1000	1000

1 grapefruit cake
125 g of flour
1 egg
1 grapefruit
300 g of sugar
150 g of butter

Cake	Flour (g)	Eggs	Grape fruit	Sugar (g)	Butter (g)
1	125	1	1	300	150
8	1000	8	8	2400	1200

b) For each cake, how many cakes can you make with 1 kg of flour and how much of each other ingredient is needed?

4 vanilla cakes (see ratio table above for other ingredients)

5 banana cakes (see ratio table above for other ingredients)

8 grapefruit cakes (see ratio table above for other ingredients)

2) Jacob has spilt chocolate sauce all over his ratio table for making chocolate brownies. Work out the missing values.

Serves (people)	Butter (g)	Cocoa (g)	Sugar (g)	Eggs	Flour	Vanilla extract (ml)
10	200	200	175	3	100	10
5	100	100	87.5	1.5	50	5
20	400	400	350	6	200	20
15	300	300	262.5	4.5	150	15

3) Identify if each statement is true or false for the following using this ratio table.

Burger	Veggie patty (g)	Lettuce (slices)	Cheese slice	Tomato slice
1	120	2	3	4

Statement	T	F
For every 4 slices of lettuce there are 6 slices of cheese.	✓	
Tomato slices are double the lettuce slices.	✓	
For every 60 g of veggie patty, there are 2 slices of tomatoes.	✓	



Task B: Using reasoning with ratios

1)

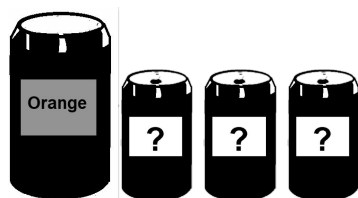
Lemons	Water	Sugar
1	500	2
2	1000	4
4	2000	8
3	1500	6

Ratio	Too sweet	Too bitter	Perfect
2 lemons, 1000 ml of water and 3 tablespoons of sugar		✓	
4 lemons, 2000 ml of water and 9 tablespoons of sugar	✓		
3 lemons, 1500 ml of water and 6 tablespoons of sugar			✓

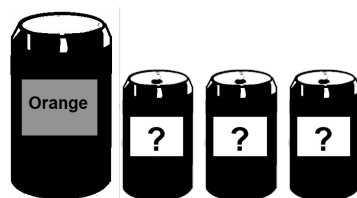
2) Orange paint is made using 2 tins of yellow for every one tin of red.

Sam, Jacob and Aisha are given an orange paint made in this ratio.

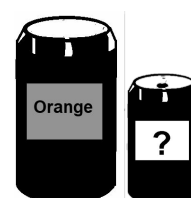
They each add three more tins to the mix, using only red and yellow.



Sam



Jacob



Aisha

The results after they mixed the extra tins are below.

What colours were in each of their three tins?

a) Sam's paint turns lighter.

The ratio of paint must include more yellow than red, and it must be a higher proportion of yellow than the original. Since there are three cans used, they must all be yellow as one red and two yellow would maintain the original colour.

b) Jacob's paint stays the same colour.

The paint stays the same colour so the paint added must be the same as the original mix, so two yellow and one red.

c) Aisha's paint turns darker.

The paint becomes more red, so either all three cans are red, or two are red and one is yellow. It's not possible to tell which.



Task B: Using reasoning with ratios

3) Tick if the statement is true or false.

Statement	True	False
A ratio describes the relationship between values.	✓	
Ratio uses an additive relationship.		✓
Adding the same amount to each part in a ratio keeps the ratio the same.		✓
Multiplying by the same amount for each part in a ratio keeps the ratio the same.	✓	

